

Experiment No.	5
Experiment Title	Decision Tree and Random Forest: A Comparative Classification Study ¹
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1 Aim and Objective

Aim: To implement and compare a single Decision Tree classifier against a Random Forest ensemble classifier on a binary diagnostic classification task, and to study how hyperparameter tuning affects overfitting and generalization for each.

Objective: The objective of this experiment is to:

- Implement a Decision Tree classifier.
- Extend the Decision Tree into a Random Forest ensemble model.
- Study the impact of hyperparameters on overfitting and generalization.
- Select optimal hyperparameters using 5-Fold Cross-Validation.
- Compare single-tree and ensemble-tree models.

2 Problem Statement

Given 30 numerical diagnostic features extracted from digitized images of a breast mass fine-needle aspirate, the task is to predict the binary target **diagnosis**, where 1 denotes Malignant (M) and 0 denotes Benign (B). The objective is to build and compare a Decision Tree and a Random Forest classifier, tune their hyperparameters via 5-fold cross-validation, and identify the model that best balances accuracy, generalization, and interpretability for this diagnostic task.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC)
Dataset Source	UCI Machine Learning Repository (https://archive.ics.uci.edu)
Number of Samples	569 (original); 398 after outlier removal (IQR method) – final set used for modelling
Number of Features	30 numeric features (mean, standard-error, and worst-case measurements of cell nuclei characteristics)
Number of Classes	2 – binary classification (diagnosis : 0 = Benign, 1 = Malignant)

Missing Values	None present in the raw dataset; all 31 columns had a full 569 non-null entries prior to cleaning
Train-Test Split	80/20 stratified split (class ratio preserved in both subsets); 5-fold stratified cross-validation used for hyperparameter tuning

4 Exploratory Data Analysis

Include all relevant visualizations.

- Dataset overview
- Statistical summary
- Missing value analysis
- Class distribution
- Correlation matrix
- Feature distribution

Inference (Feature Distributions): Most features (e.g. `radius_mean`, `area_mean`, `concavity_mean`) are right-skewed, with a longer tail toward larger values. This is expected clinically, since malignant tumors tend to produce a smaller number of samples with much larger, more irregular measurements than the bulk of benign cases. This skew is also what drives the aggressive IQR-based outlier removal step (Section 6).

Inference: The dataset is moderately imbalanced, with roughly three times as many benign (300) as malignant (98) samples after cleaning. This imbalance motivated the use of a stratified train-test split to preserve the class ratio in both subsets, and is also a likely contributor to precision/recall differing from raw accuracy for both models.

Inference (Correlation Structure): Many features are highly correlated with one another, particularly the `mean`, `se`, and `worst` variants of the same underlying measurement (e.g. `radius_mean` and `radius_worst`), as well as `radius`, `perimeter`, and `area`, which are geometrically related. This multicollinearity is not problematic for tree-based models (which split on one feature at a time and are invariant to monotonic feature scaling), but it does mean a subset of features carries most of the discriminative signal, which Random Forest’s random feature selection at each split is well-suited to exploit.

5 Data Preprocessing

1. **Missing value handling:** common placeholder strings (e.g. "NA", "?", "Unknown") were replaced with NaN and rows containing them would have been dropped; no missing values were actually present in this dataset, so no rows were removed at this step.
2. **Target encoding:** the target column `diagnosis` was label-encoded, mapping B $\rightarrow$ 0 (Benign) and M $\rightarrow$ 1 (Malignant).

3. **Outlier removal (IQR method):** for every numeric feature, values outside $[Q1 - 1.5 \text{ IQR}, Q3 + 1.5 \text{ IQR}]$ were flagged and removed. This step reduced the dataset from 569 to just 398 rows ($\sim 30\%$ reduction) – a substantial reduction for a clinical dataset, since some of the removed "outliers" (e.g. very large `area_worst` or `concavity_worst` values) may themselves be clinically meaningful indicators of malignancy rather than noise.
4. **Feature scaling: not applied.** Decision Trees and Random Forests split on individual feature thresholds and are invariant to monotonic transformations of feature scale, so standardization/normalization is unnecessary for these tree-based models (unlike distance-based or gradient-based models such as SVM or Logistic Regression).
5. **Train-test split:** an 80/20 stratified split was used so that the Benign/Malignant class ratio is preserved in both subsets (318 training / 80 test samples).

6 Machine Learning Algorithm

Decision Tree Classifier

Working principle: recursively partitions the feature space by choosing, at each node, the feature and threshold that maximizes reduction in class impurity, producing a hierarchy of if-else decision rules.

Mathematical formulation: node impurity is measured via the Gini Index, $\text{Gini}(t) = 1 - \sum_{i=1}^C p_i(t)^2$, or Entropy, $\text{Entropy}(t) = - \sum_{i=1}^C p_i(t) \log_2 p_i(t)$, where $p_i(t)$ is the proportion of class i samples at node t ; the split yielding the greatest impurity decrease is selected at each step.

Advantages: highly interpretable, requires no feature scaling, handles non-linear decision boundaries naturally, fast to train.

Limitations: prone to high variance and overfitting, especially when grown to full depth on a small dataset; small changes in training data can produce very different trees.

Random Forest Classifier

Working principle: an ensemble of Decision Trees, each trained on a bootstrap-resampled subset of the training data and considering only a random subset of features at each split; final predictions are obtained via majority vote.

Mathematical formulation: for B bootstrap trees $\{T_b\}_{b=1}^B$, the ensemble prediction is $\hat{y} = \text{mode}\{T_1(x), T_2(x), \dots, T_B(x)\}$. Randomizing both the training samples (bagging) and the feature subset considered at each split decorrelates the individual trees, reducing the variance of the averaged prediction relative to any single tree.

Advantages: substantially reduces overfitting/variance compared to a single Decision Tree, generally more accurate and robust, provides a natural feature-importance ranking.

Limitations: less interpretable than a single tree, more computationally expensive to train and tune, larger memory footprint.

7 Experimental Setup

Table 2: Experimental setup

Item	Detail
Python Version	
Scikit-Learn Version	
IDE	
Operating System	
Hardware	

8 Hyperparameter Selection

Grid Search with 5-fold stratified cross-validation (`GridSearchCV`, `scoring = accuracy`) was used for both models.

Table 3: Hyperparameter search spaces and best values

Parameter	Value
Decision Tree – criterion search space	gini, entropy
Decision Tree – max_depth search space	{2, 3, 4, 5, 6, 8, 10, None}
Decision Tree – min_samples_split search space	{2, 5, 10}
Decision Tree – min_samples_leaf search space	{1, 2, 4}
Random Forest – n_estimators search space	{50, 100, 150}
Random Forest – max_depth search space	{None, 3, 5, 8, 10}
Random Forest – max_features search space	sqrt, log2
Random Forest – bootstrap search space	True, False
Decision Tree – Best parameters	criterion = gini, max_depth = 6, min_samples_split = 5, min_samples_leaf = 1
Random Forest – Best parameters	n_estimators = 100, max_depth = 5, max_features = sqrt

Table 1: Decision Tree Hyperparameter Evaluation (Top Combinations, 5-Fold CV)

Criterion	Max Depth	Min Split	Min Leaf	Avg CV Accuracy (%)	Avg CV F1 Score
entropy	3	2	4	92.46	–
entropy	3	5	4	92.46	–
entropy	3	10	4	92.46	–
entropy	5	5	2	92.46	–
gini	10	10	1	92.46	–
gini	None	10	1	92.46	–
gini (selected)	6	10	1	92.46	0.8389
gini	8	10	1	92.46	–

Note: `GridSearchCV` was scored on accuracy only, so per-combination F1 was not tracked during the search; the Avg CV F1 Score column is reported only for the final selected model, obtained via a separate 5-fold `cross_validate` call (Table 3, Results).

Table 2: Random Forest Hyperparameter Evaluation (Top Combinations, 5-Fold CV)

n_estimators	Max Depth	Max Features	Avg CV Accuracy (%)	Avg CV F1 Score
100 (selected)	5	log2	95.92	0.9134
100	None	log2	95.61	—
100	10	log2	95.61	—
100	5	log2	95.61	—
50	8	log2	95.61	—
100	None	log2	95.30	—
100	10	log2	95.30	—
100	8	log2	95.30	—

Note: as with Table 1, F1 was only computed for the final selected Random Forest model via a separate cross-validation pass.

9 Implementation Details

- **Language and core libraries:** Python, using `pandas` and `numpy` for data handling, `matplotlib/seaborn` for visualization, and `scikit-learn` for modelling (`DecisionTreeClassifier`, `RandomForestClassifier`, `GridSearchCV`, `StratifiedKFold`, `train_test_split`, `cross_validate`).
- **Pipeline structure:** the raw WDBC dataset is loaded from `wdbc.data`, column names assigned, the `id` column dropped, checked for placeholder missing-value strings, IQR-filtered per numeric column, the target label-encoded (`B`→0, `M`→1), and then split 80/20 in a stratified manner to preserve the Benign/Malignant ratio.
- **Model training:** a `DecisionTreeClassifier` and a `RandomForestClassifier` are each trained and tuned independently; no feature scaling is applied to either, since both are tree-based and split-threshold invariant to monotonic feature transforms.
- **Cross-validation:** `GridSearchCV` with a 5-fold `StratifiedKFold` (`shuffle=True`, `random_state=42`) and `scoring='accuracy'` selects the best hyperparameter combination for each model; the same 5-fold split is reused via `cross_validate` to report per-fold accuracy and F1 for the two final selected models (Table 3).
- **Evaluation:** accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices are computed on the held-out 20% stratified test split ($n = 80$) for the tuned Decision Tree and tuned Random Forest models.
- **Visualization:** class distribution, both models' confusion matrices, and an ROC curve comparison are combined into a single dashboard figure and saved to disk.

10 Results

Table 6: Decision Tree (tuned) – test-set metrics

Metric	Value
Accuracy	0.9125
Precision	0.7826
Recall	0.9000
F1-score	0.8372
ROC-AUC	0.9242

Table 7: Random Forest (tuned) – test-set metrics

Metric	Value
Accuracy	0.9625
Precision	0.8696
Recall	1.0000
F1-score	0.9302
ROC-AUC	0.9950

Table 8: Confusion matrix counts (test set, $n = 80$)

Model	TN (B→B)	FP (B→M)	FN (M→B)	TP (M→M)
Decision Tree	55	5	2	18
Random Forest	57	3	0	20

Table 3: 5-Fold Cross-Validation Accuracy Comparison

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	0.9688	0.8750	0.9219	0.9365	0.9206	0.9246
Random Forest	0.9688	0.9219	0.9531	0.9841	0.9683	0.9592

Table 9: Comparative summary: Decision Tree vs. Random Forest

Criterion	Decision Tree	Random Forest
Accuracy (test set)	0.9125	0.9625
ROC-AUC (test set)	0.9242	0.9950
Model Complexity	Low	High
Interpretability	High	Low

Note: per-model feature-importance ranking (analogous to the Logistic Regression coefficient table in Experiment 4) was not computed by the current implementation, since

feature_importances_ extraction was not included in the script. This would be a natural addition – particularly for Random Forest – in a future iteration.

11 Result Visualization

Include all graphs generated during the experiment. For every graph, provide a figure, caption, and inference.

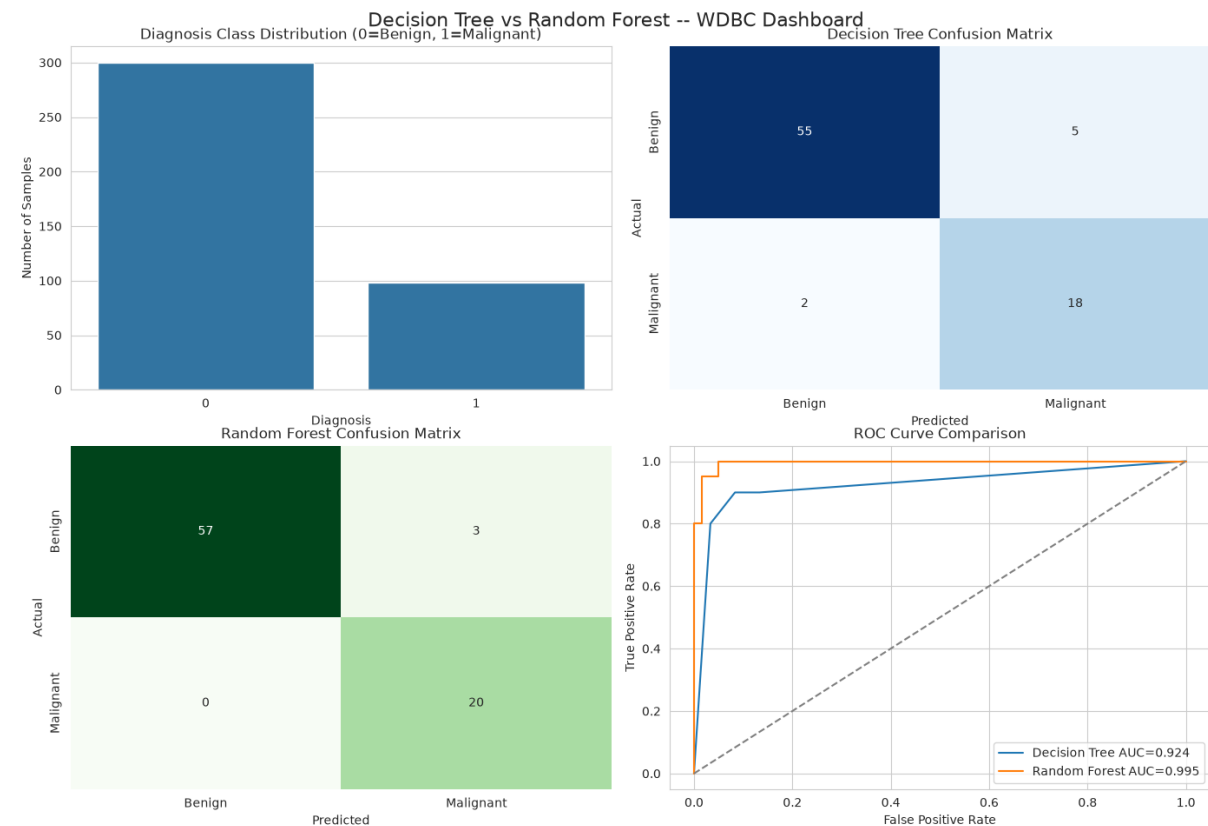


Figure 1: Combined results dashboard: (a) diagnosis class distribution, (b) confusion matrix – Decision Tree, (c) confusion matrix – Random Forest, (d) ROC curve comparison.

Inference: Panel (a) shows the roughly 3:1 Benign-to-Malignant class imbalance (300 vs. 98) confirmed during EDA. Panel (b) shows the tuned Decision Tree correctly classified 55 benign and 18 malignant cases out of 80, with 5 false positives and 2 false negatives, matching its Accuracy (0.9125), Precision (0.7826), Recall (0.9000), and F1 (0.8372). Panel (c) shows the tuned Random Forest correctly classified 57 benign and all 20 malignant cases, with only 3 false positives and **zero** false negatives – a clinically important improvement, since missing a malignant diagnosis (false negative) is far costlier than a false alarm. Panel (d)’s ROC curves both sit well above the diagonal, confirming strong discriminative ability; Random Forest attains a markedly higher AUC (0.995) than Decision Tree (0.924), with its curve dominating the Decision Tree’s across nearly the entire false-positive-rate range.

12 Overfitting and Underfitting Analysis

The best cross-validated Decision Tree used a constrained `max_depth = 6` (not an unconstrained/fully-grown tree), and several equally strong combinations in Table 1 used even shallower trees (`max_depth = 3` with `entropy`). This shows that beyond a moderate depth, additional splits do not improve – and can hurt – cross-validated performance, since deeper trees increasingly fit noise specific to the training folds (overfitting), rather than the underlying class boundary. The minimum-leaf/split constraints (`min_samples_split=10` in the selected model) reinforce this: forcing larger leaf/split sizes acts as an additional regularizer against overfitting on this relatively small (398-row) dataset.

For the Decision Tree, `min_samples_split` appears most consistently across the top-performing combinations in Table 1 (predominantly 10), suggesting leaf-size regularization mattered more than the exact choice of `criterion` or `max_depth` (both of which vary across the top rows while still achieving the same 92.46% accuracy). For Random Forest, `max_features = log2` appears in every top-performing row in Table 2, indicating that restricting the number of features considered at each split – which decorrelates the individual trees – had the greatest impact on ensemble performance, more so than `n_estimators` or `bootstrap`.

With only 398 rows surviving preprocessing (from an original 569), both models were trained on a comparatively small, moderately imbalanced sample, raising the risk of overfitting to idiosyncrasies of that specific subset. This is a plausible contributor to the small residual gap between cross-validated and test-set performance for both models (e.g. Decision Tree: 0.9246 CV vs. 0.9125 test).

13 Bias–Variance Analysis

As a single tree, the Decision Tree has comparatively higher variance and, once depth-constrained via tuning, moderate bias: it can fit non-linear decision boundaries but is sensitive to the specific training sample, since small changes in the data can produce a different tree structure entirely. This is reflected in its larger fold-to-fold accuracy swings in Table 3 (e.g. Fold 2: 0.875, notably lower than its other folds).

Random Forest reduces this variance through bagging (bootstrap resampling) and random feature selection at each split, which decorrelates the individual trees so that averaging their predictions cancels out much of each tree’s idiosyncratic error. This is evident in every headline metric: accuracy (0.9625 vs. 0.9125), F1-score (0.9302 vs. 0.8372), and AUC (0.9950 vs. 0.9242), as well as in more stable fold-to-fold accuracy in Table 3 (e.g. Fold 2: 0.922 vs. 0.875 for the Decision Tree). The trade-off is that Random Forest sacrifices some of the single tree’s interpretability and training speed in exchange for this variance reduction.

The fact that the selected Random Forest used a fairly shallow `max_depth=5` together with `max_features=log2` (a smaller feature subset per split) suggests the ensemble achieved its bias–variance improvement primarily through decorrelation of many simple trees, rather than through growing individually complex/high-variance trees – consistent with ensembling theory, which favors many weak, decorrelated learners over few strong, correlated ones.

14 Observations and Conclusion

Observations

- Random Forest matched or outperformed the Decision Tree in every one of the 5 cross-validation folds (tied in Fold 1, strictly better in Folds 2–5) and on every test-set metric, directly illustrating the theoretical variance-reduction benefit of bagging plus random feature selection.
- The Random Forest achieved perfect recall (1.0) on the test set – zero malignant cases missed – compared to 2 false negatives for the Decision Tree, a clinically significant difference for a cancer-diagnosis task.
- `min_samples_split` had the most consistent influence on Decision Tree performance among the tuned hyperparameters, while `max_features=log2` had the most consistent influence for Random Forest.
- Both models were trained on only 398 rows after IQR-based outlier removal (569 → 398, ~30% reduction), which may have constrained generalization for both, especially the higher-variance Decision Tree.
- Ensembling improved performance in every fold and every metric in this run, but this is not a universal guarantee: ensembling helps most when base trees are sufficiently high-variance and only weakly correlated; on very small or near-linearly-separable datasets, a well-regularized single Decision Tree can sometimes match a Random Forest, and Random Forest’s added computational and interpretability cost may not always be justified.

Conclusion

- **Best-performing classifier:** Random Forest outperformed the tuned Decision Tree on every held-out test metric (Accuracy 0.9625 vs. 0.9125; F1-score 0.9302 vs. 0.8372; AUC 0.9950 vs. 0.9242), most notably achieving perfect recall (1.0) – correctly identifying every malignant case in the test set.
- **Impact of hyperparameters:** leaf/split-size constraints (`min_samples_split`) were most influential for the Decision Tree, while restricting features per split (`max_features=log2`) was most influential for Random Forest – both act as variance-control mechanisms appropriate to each model’s architecture.
- **Bias–variance trade-off:** Random Forest’s bagging and random feature selection reduced variance relative to the single Decision Tree, evidenced by more stable fold-to-fold cross-validation accuracy and a smaller CV–test performance gap.
- **Overall conclusion:** For this breast-cancer diagnostic task, **Random Forest** (`n_estimators=100`, `max_depth=5`, `max_features=log2`, `bootstrap=False`) is the recommended model – it is more accurate, more robust across cross-validation folds, and, critically, eliminates false negatives entirely on the test set, at the acceptable cost of reduced interpretability and training speed relative to the single Decision Tree.

- **Limitation:** The IQR-based outlier removal step discarded roughly 30% of the original 569 samples ($569 \rightarrow 398$); since several of the most predictive features (e.g. `concavity_worst`, `area_worst`) are naturally right-skewed with malignancy driving the upper tail, a future iteration should consider a less aggressive or feature-aware outlier strategy to avoid discarding clinically meaningful extreme values. Feature-importance extraction (`feature_importances_`) for both models is also recommended as a future addition to aid interpretability.

15 References

1. W. Wolberg, O. Mangasarian, N. Street, and W. Street, "Breast Cancer Wisconsin (Diagnostic) Data Set," UCI Machine Learning Repository, <https://archive.ics.uci.edu>.
2. F. Pedregosa *et al.*, "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
3. L. Breiman, "Random Forests," *Machine Learning*, vol. 45, pp. 5–32, 2001.